

21-52 - Q4

Q: (a) filter
wrapper

diagrams

difference

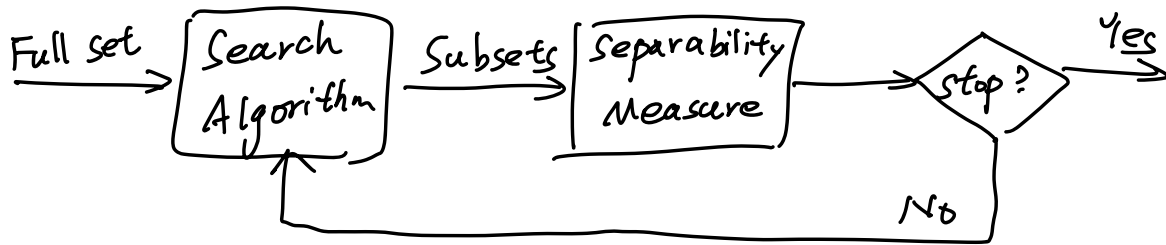
advantages

disadvantages

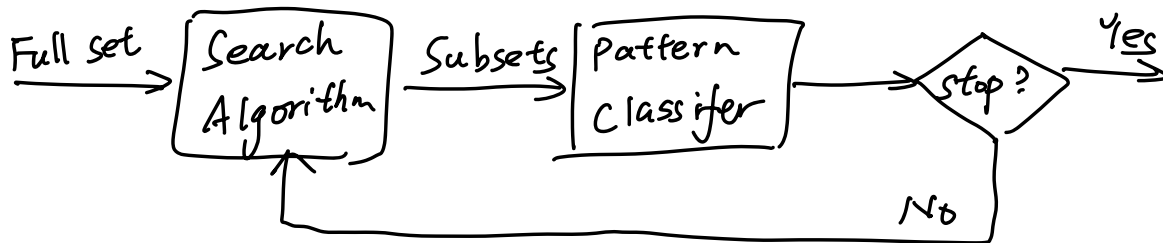
(b) clustering
example.
Scenario

(c) DBSCAN
procedures

Solution ca) ① filter



② wrapper



③ Difference

(1) Filter methods does not include a classifier in the loop, instead, it employs class separability measures as the feature subset evaluation criterion

(2) The wrapper method include a classifier in the loop and use classification performance such as accuracy or error rate as the feature subset evaluation criterion

④ Filter

Adv. ① computationally fast

② simple to implement

③ don't depend on a specific learning algorithm, model agnostic

④ Scalable to very high-dimensional data

Disadv. ① less accurate

② don't account for features interactions

③ may discard features that individually have small correlation with y but together could be highly informative

⑤ Wrapper

Adv. ① account for features interactions

② better performance

③ find more efficient feature

Disadv. ① computationally expensive

② easily overfit

(b) ① (1) Centroid - Based Clustering

(2) K - means

(3) roughly know num of clusters

relatively spherical, similarly sized clusters

② (1) Hierarchical Clustering

(2) Agglomerative algorithm

divisive algorithm

(3) need hierarchy cluster rather than partition

③ (1) Density - Based Clustering

(2) DBSCAN

Density - Based Spatial Clustering
of Applications with Noise

(3) arbitrary shape data sets

data contain noise/outliers

(c) ① set $MinPts$ and ϵ

randomly pick a data point as starting point

② find recursively all its density connected points and assign them to the same cluster as the core point

③ Randomly choose another point that haven't been visited and go to step-2

④ process is finished when all points are visited

Those points that don't belong to any cluster are considered as noise